

JACOB LAWSON

310-971-0520 | jacobslawson04@gmail.com | [linkedin.com/in/lawso](https://www.linkedin.com/in/lawso) | github.com/L4wson

EDUCATION

University of California, Merced

Expected May 2027

B.S. in Computer Science & Engineering

Merced, CA

- **Relevant Coursework:** Database Systems, Full-Stack Web Development, Computer Architecture, Data Structures, Algorithm Design & Analysis, Linear Algebra

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS, C++, PowerShell, Go

Frameworks & Libraries: React, Flask, SQLAlchemy, Node.js, React Native, Vite, Pandas, NumPy

Tools & Platforms: PostgreSQL, Git, GitHub Actions, Linux, AWS, Docker, Vercel, Render

Concepts: REST APIs, Relational Data Modeling, Data Pipelines, Metrics Collection, LLM Integration, Auth, CI/CD

EXPERIENCE

Software Engineering Intern

May 2025 – Jul. 2025

Robotics Cats

CA

- Built a PowerShell monitoring service collecting status from 120 camera endpoints on 30–120s polling intervals into an auto-updating JSON state file, giving the team a single source of truth for endpoint health in place of manual checks.
- Developed a React Native + Flask OCR pipeline processing 40 receipts/day at 96% accuracy, auto-syncing extracted line items to Google Sheets and eliminating 120 min/day of manual entry.
- Integrated Gemini LLM endpoints into a full-stack Python/Flask + React/Vite application deployed on AWS, generating quiz content dynamically at request time.

Student Technology Consultant

Aug. 2023 – Present

UC Merced – Office of Information Technology

Merced, CA

- Resolved 80+ ServiceNow tickets across OS, networking, AV hardware, and SaaS issues in enterprise classroom systems.
- Documented recurring failure patterns in a Confluence runbook that became the team's first stop before escalating.

Undergraduate Research Assistant

Oct. 2022 – May 2023

UC Merced – LInC Lab

Merced, CA

- Automated cleaning of 600+ CSV files with a Python/pandas pipeline, cutting processing time 30% and retrieval time 50% across 5+ studies.
- Standardized data organization protocols across 4 research projects, reducing formatting inconsistencies 20% and simplifying handoffs between researchers.

PROJECTS

Bobcat Bulletin | *Python, BeautifulSoup, Flask, React, PostgreSQL, GitHub Actions, Vercel*

- Built a full-stack campus safety app on a scheduled GitHub Actions scraping pipeline with change detection, normalizing 8,700+ UCM PD incident records into PostgreSQL for historical trend analysis.
- Turned a hard-to-navigate institutional page into a searchable, filterable incident feed and location-based map view.

LiftLabs | *React, Vite, Flask, SQLAlchemy, PostgreSQL, REST API, Vercel, Render*

- Designed a 15+ endpoint REST API and 7-model relational schema backing a fitness forum with JWT auth, role-based moderation, and one-vote-per-user enforcement validated server-side.
- Deployed a production SPA across Vercel and Render, resolving cross-origin auth and persistent file storage issues.

Greenview | *JavaScript, Chrome Extension API, Manifest V3, HTML/CSS*

- Shipped a Chrome/Firefox extension tagging 1000+ job listings by applicant tracking system, using MutationObserver for async DOM detection and synced storage for cross-session filters.

OpenGL Paint Application | *C++, OpenGL*

- Engineered a 2D paint application in C++/OpenGL using dynamic memory management and object-oriented design, cutting render lag 15% over the initial implementation.